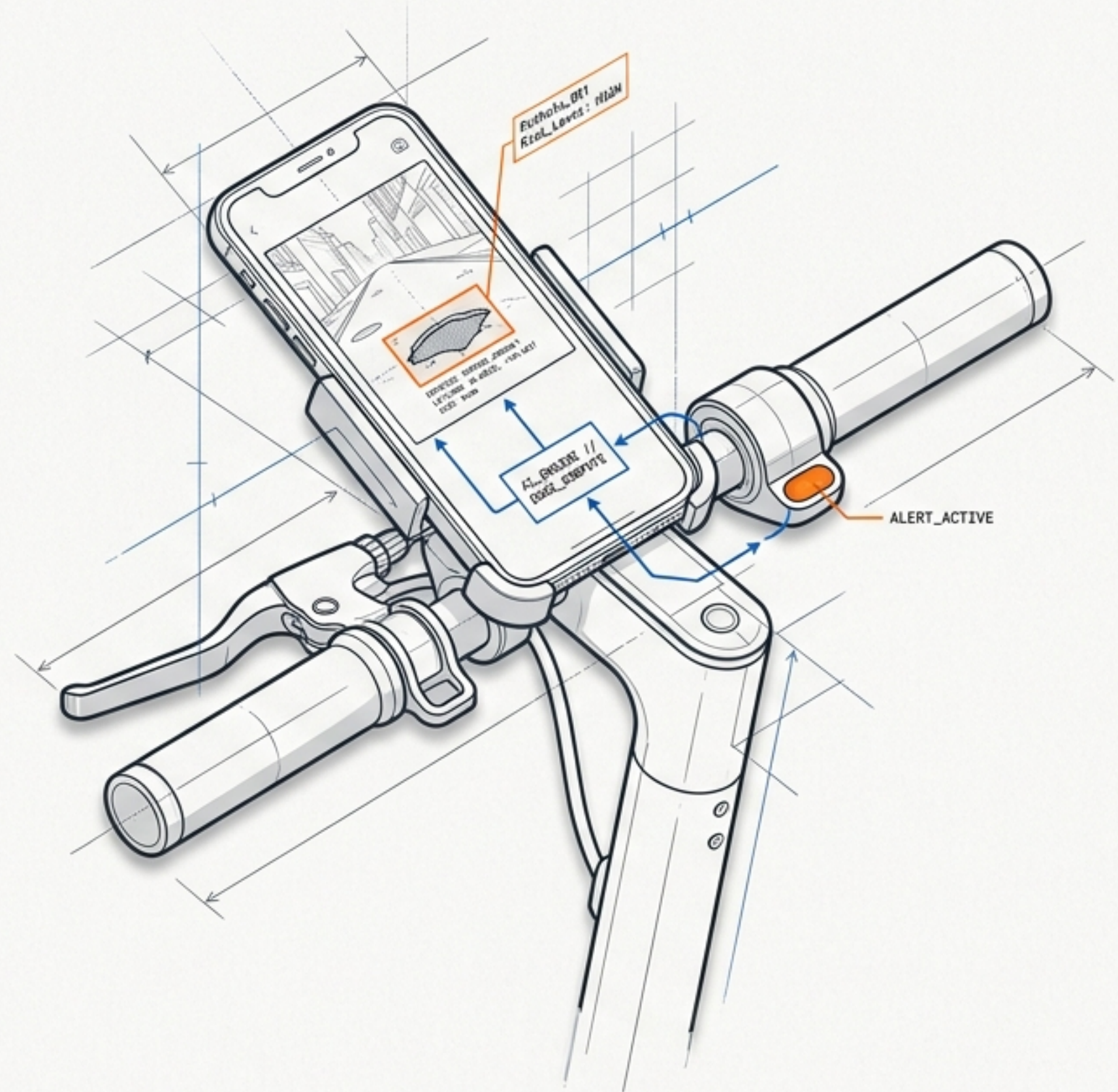


Proactive Vision for Micro-Mobility

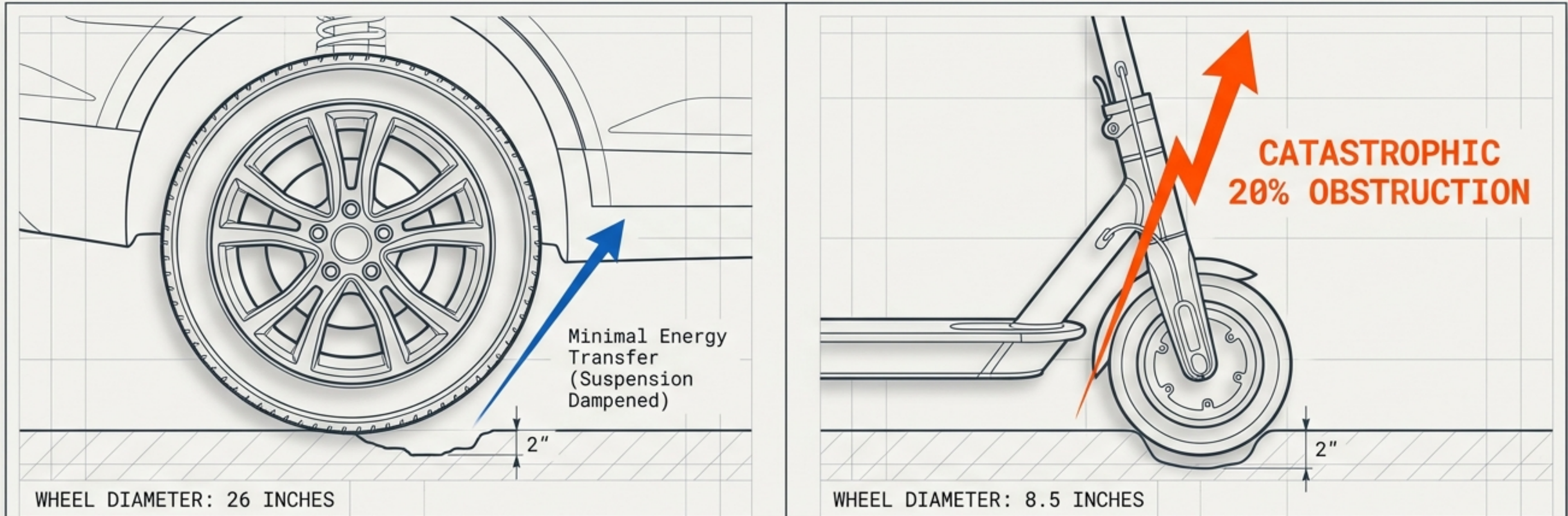
AI-Powered Real-Time
Hazard Detection & Risk
Assessment at the Edge

FULLY OFFLINE | ZERO LATENCY | PRIVACY FIRST



Micro-mobility hardware cannot absorb macro-infrastructure defects.

Safety technology must shift from impact mitigation to pre-impact avoidance.



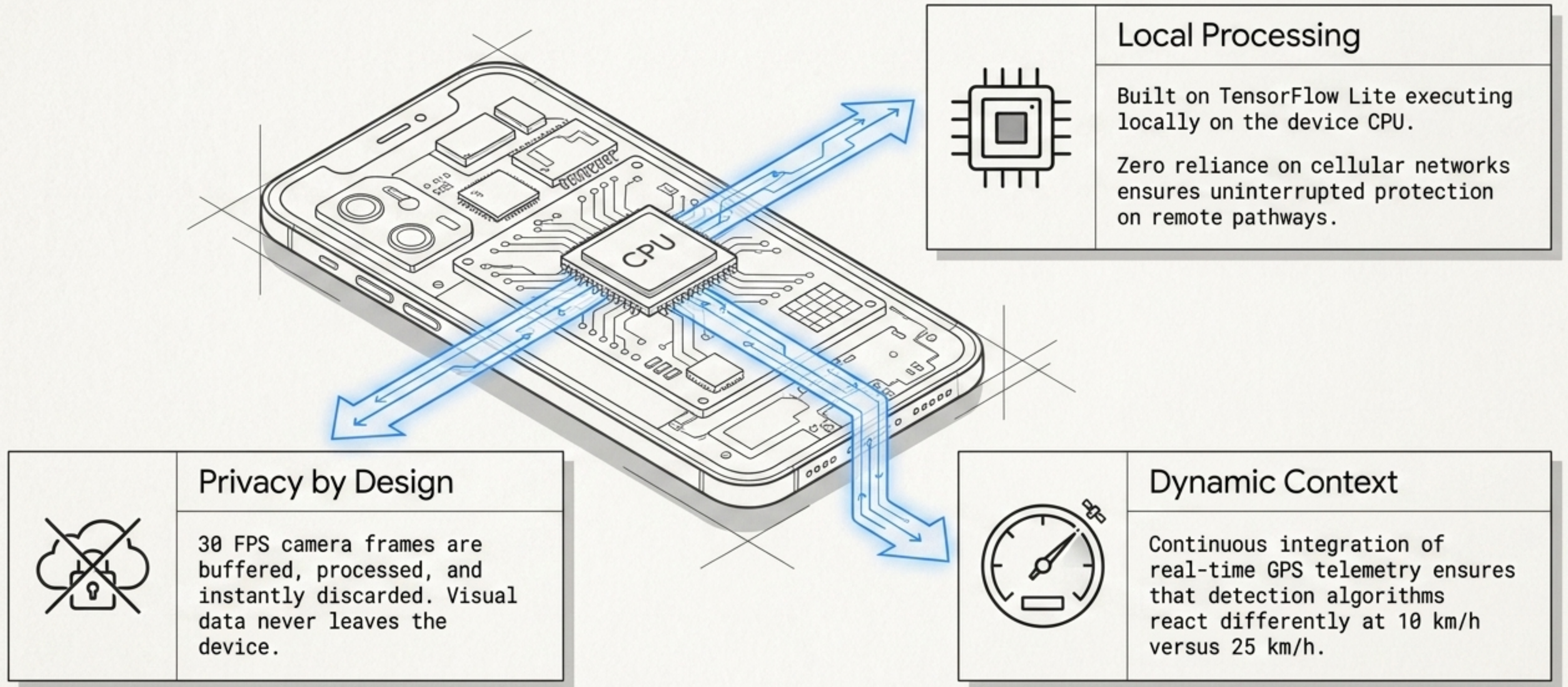
The rapid expansion of electric scooters has outpaced road maintenance. Lacking passive damping systems, kinetic energy from surface anomalies transfers directly to the rigid handlebars, causing front-wheel lockups.

The gap in urban safety technology.

	Accelerometer Systems	LiDAR Systems	Municipal Surveys	Proposed Edge App
Detection Type	Reactive (Post-impact)	Proactive (Pre-impact)	Proactive	Proactive (Pre-impact)
Cost	Low Cost	Very High Cost	High Cost	Low Cost
Edge Capabilities	Offline Log	Local/Offline	Offline Upload	Local TFLite Processing
Warning Latency	No Warning	Medium	Static Reports	<200ms

Takeaway: Only an edge-native smartphone application can deliver proactive, pre-impact warnings at a consumer-friendly deployment scale.

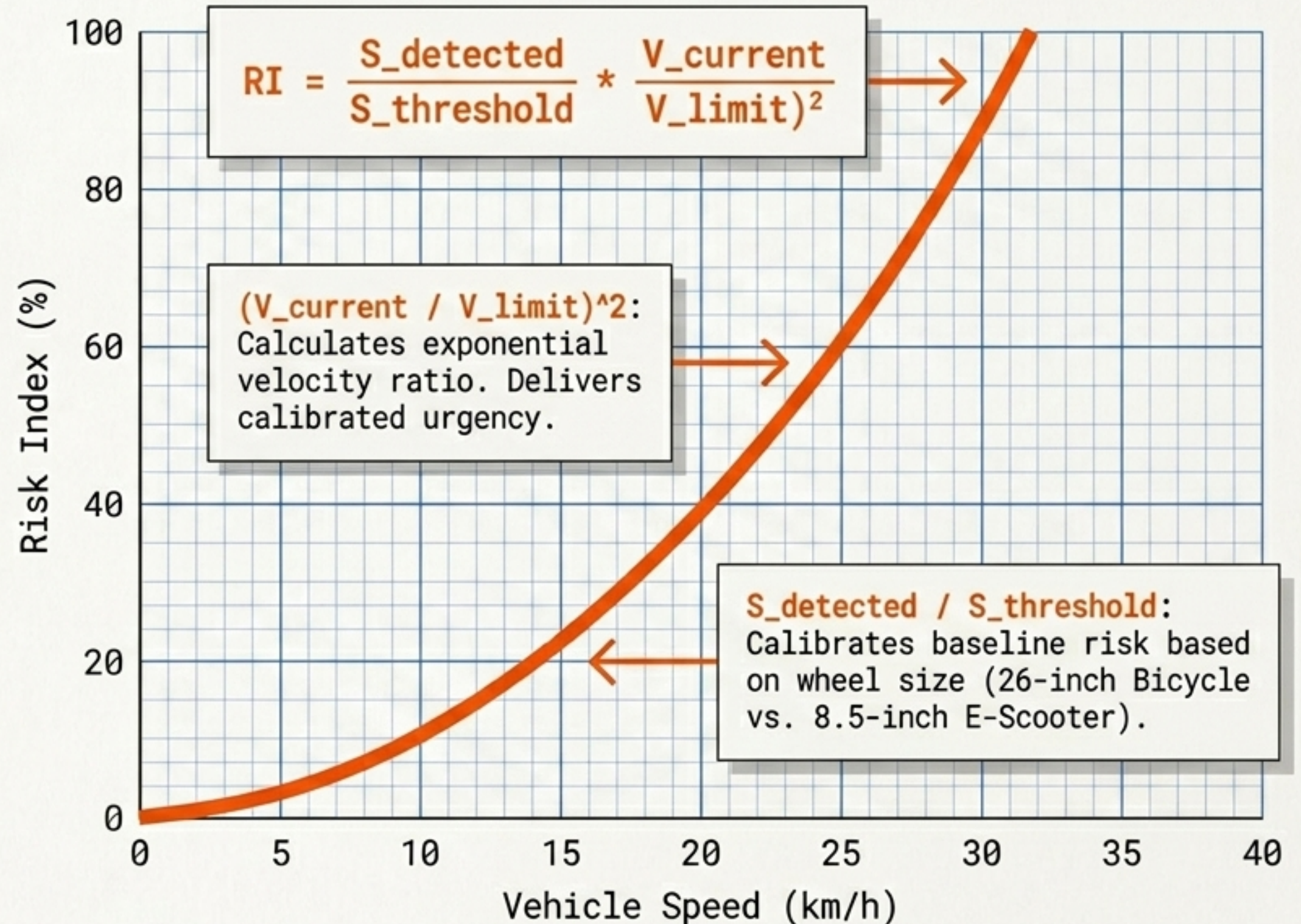
A paradigm shift powered by edge-native intelligence.



The math of danger: Dynamic risk assessment.

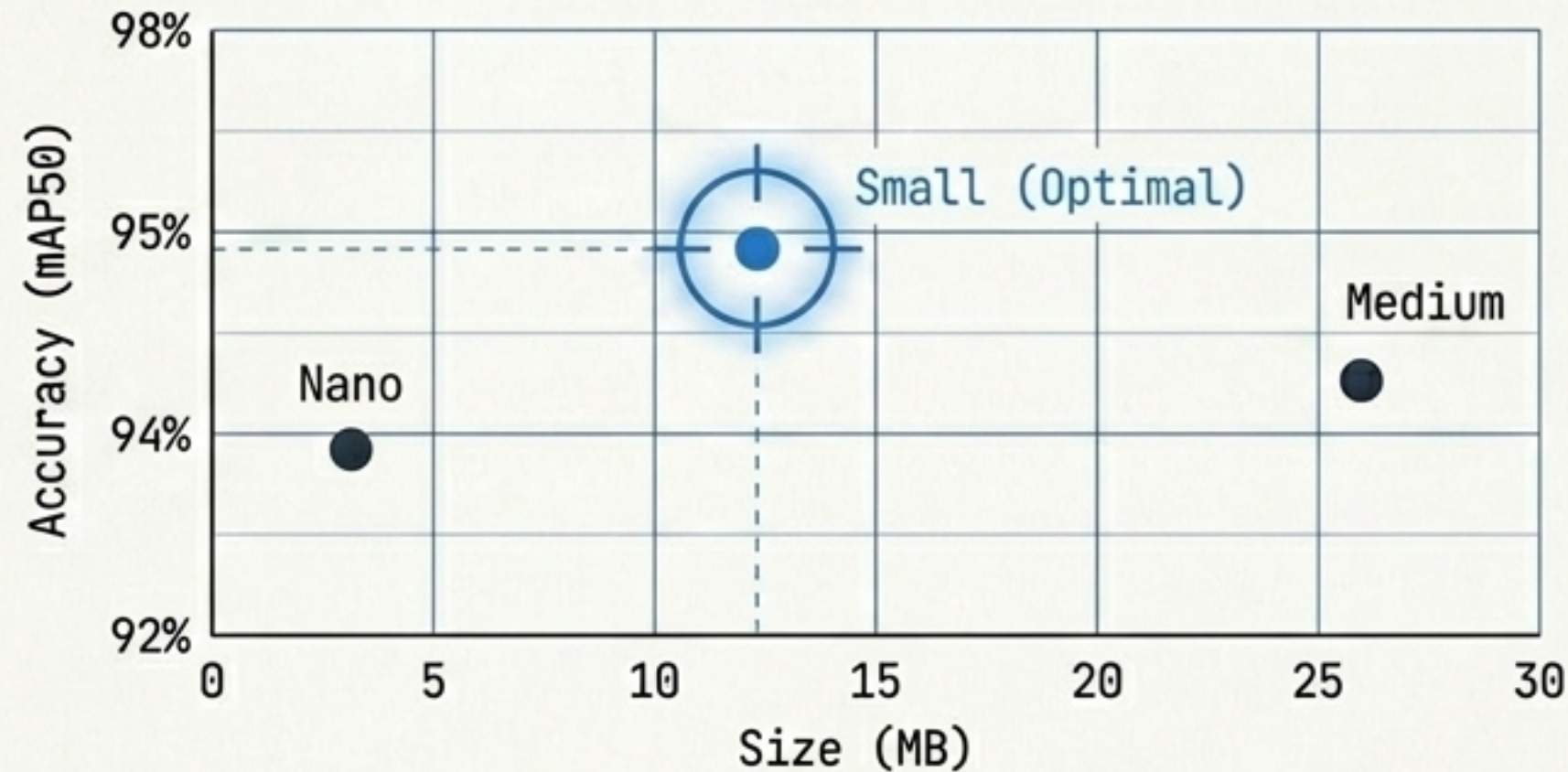
A pothole's threat level is not static. Mirroring kinetic energy physics, our Risk Assessment Algorithm scales the hazard severity exponentially based on the rider's instantaneous speed.

- Hitting a defect at 20 km/h generates **four times** the destructive energy as hitting it at 10 km/h.



Finding the edge-computing sweet spot.

Why YOLOv8 Small out-performs heavier variants on focused datasets.



YOLOv8 Nano

3.5MB Footprint | 93.9% mAP50 | 3.2M Params

Fastest, but lowest accuracy.

YOLOv8 Small (The Winner)

12.2MB Footprint | 95.8% mAP50 | 11.2M Params

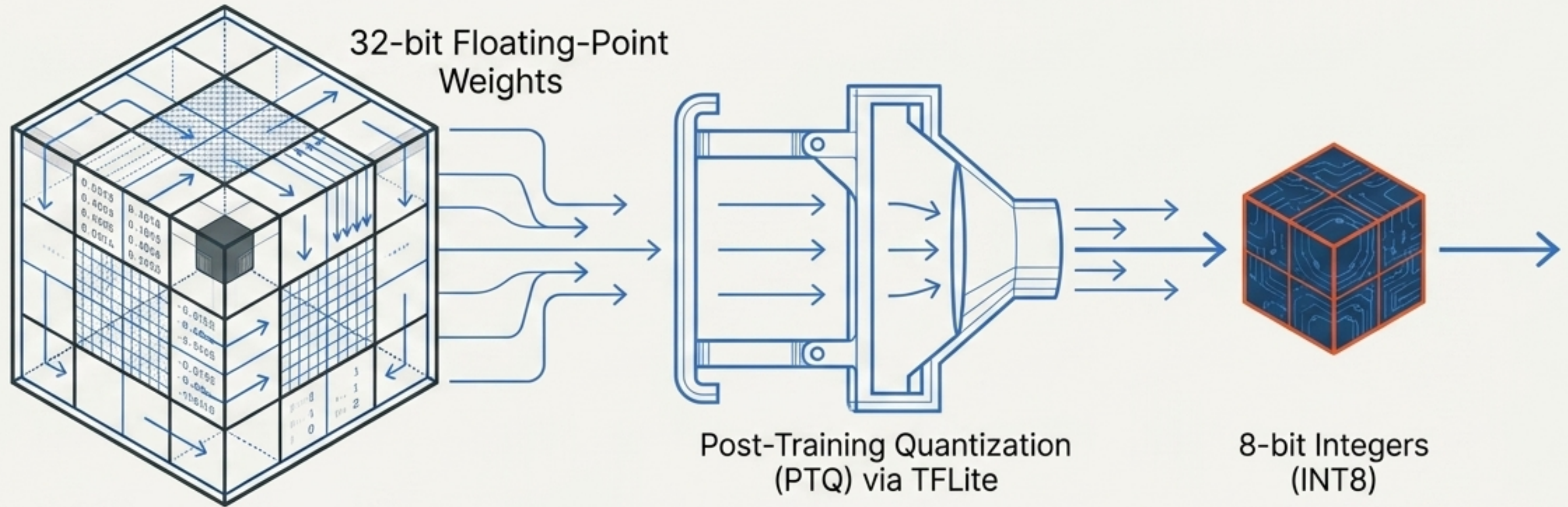
Highest accuracy. The 11.2M parameter capacity acts as an **implicit regularizer**, preventing **overfitting** on **single-class** datasets.

YOLOv8 Medium

26.2MB Footprint | 94.5% mAP50 | 25.9M Params

Highly robust, but too heavy for budget CPUs and prone to minor overfitting.

8-bit quantization ensures high-speed inference without battery drain.



Deploying an 11-million parameter neural network on a handlebar requires extreme optimization. Mapping 32-bit weights down to 8-bit integers drastically reduces physical size.

12.2 MB

Model footprint after compression

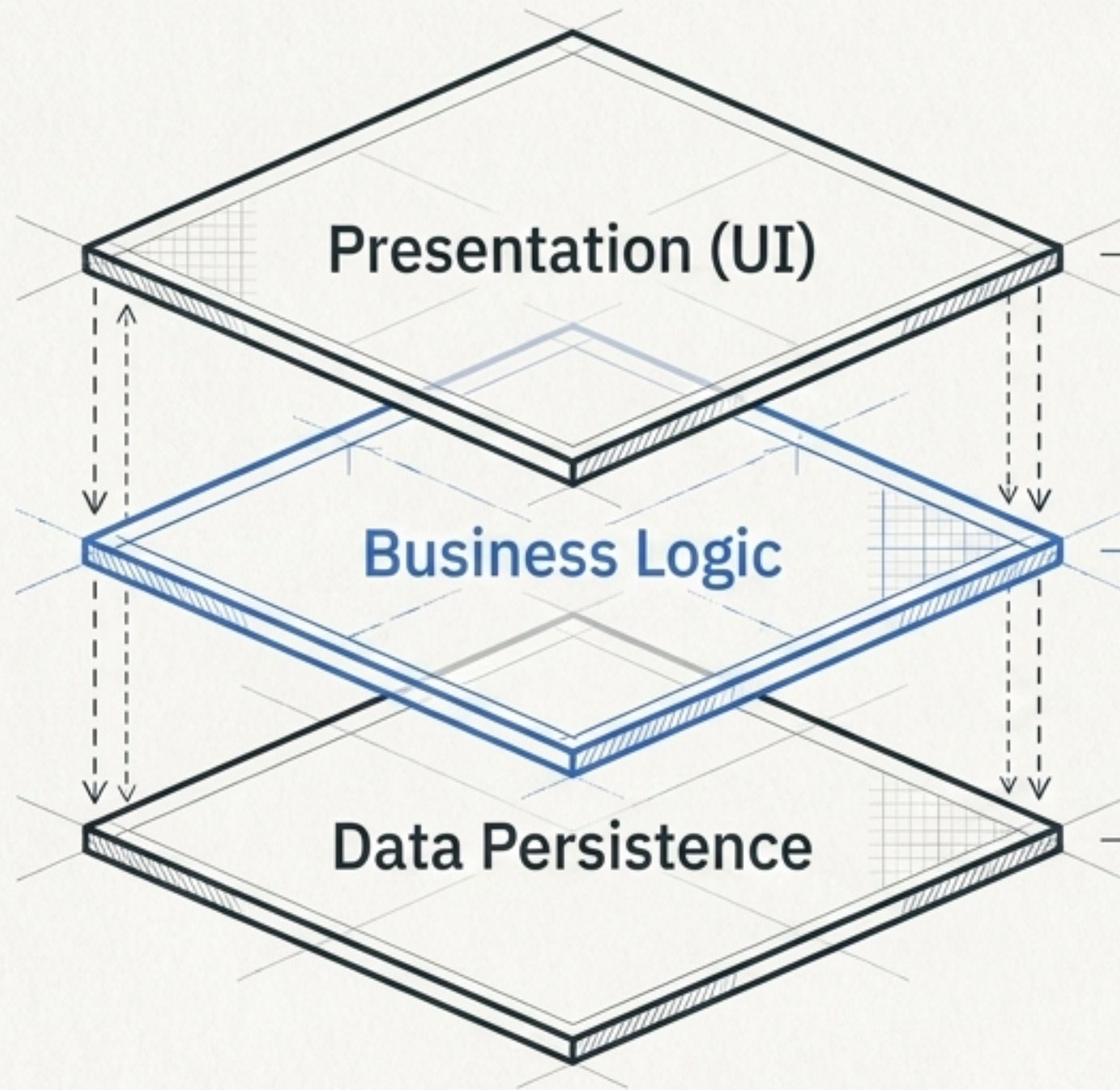
< 50 ms

Inference execution time
per frame

95.8%

Maintained mAP50 score despite aggressive quantization

Asynchronous architecture built for uninterrupted buffering.



Presentation (UI): Manages the Camera Stream Viewer, High-Contrast Alert Logic, and Map Rendering.

Business Logic: Houses the TFLite Interpreter, Geolocator API (1 Hz), and Risk Index Algorithm. Executes via background isolates to prevent UI lag.

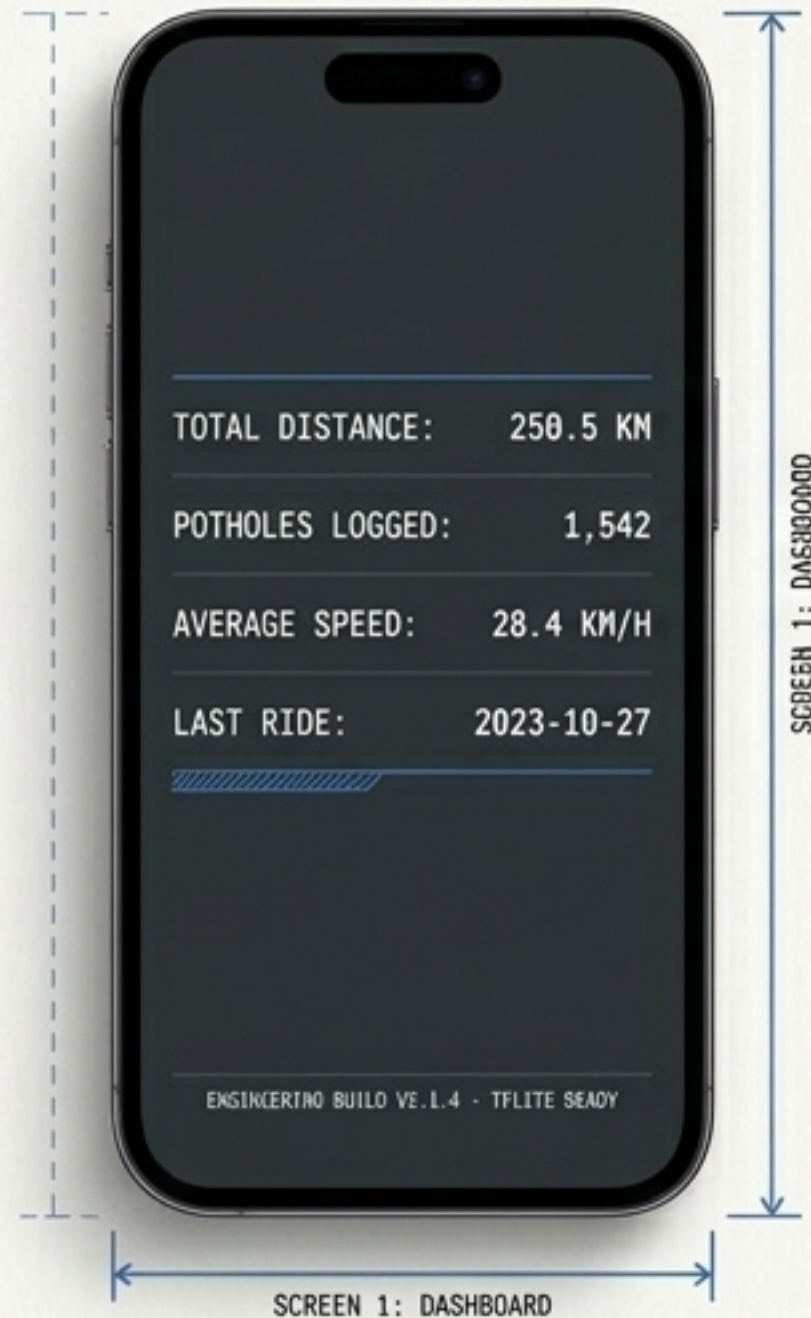
Data: Local SQLite, rides.json (stores polylines and hazard markers), and settings.json.

High-speed camera frame processing (30 FPS) is fully **decoupled** from the GPS listeners and user interface, guaranteeing a perfectly smooth HUD experience.

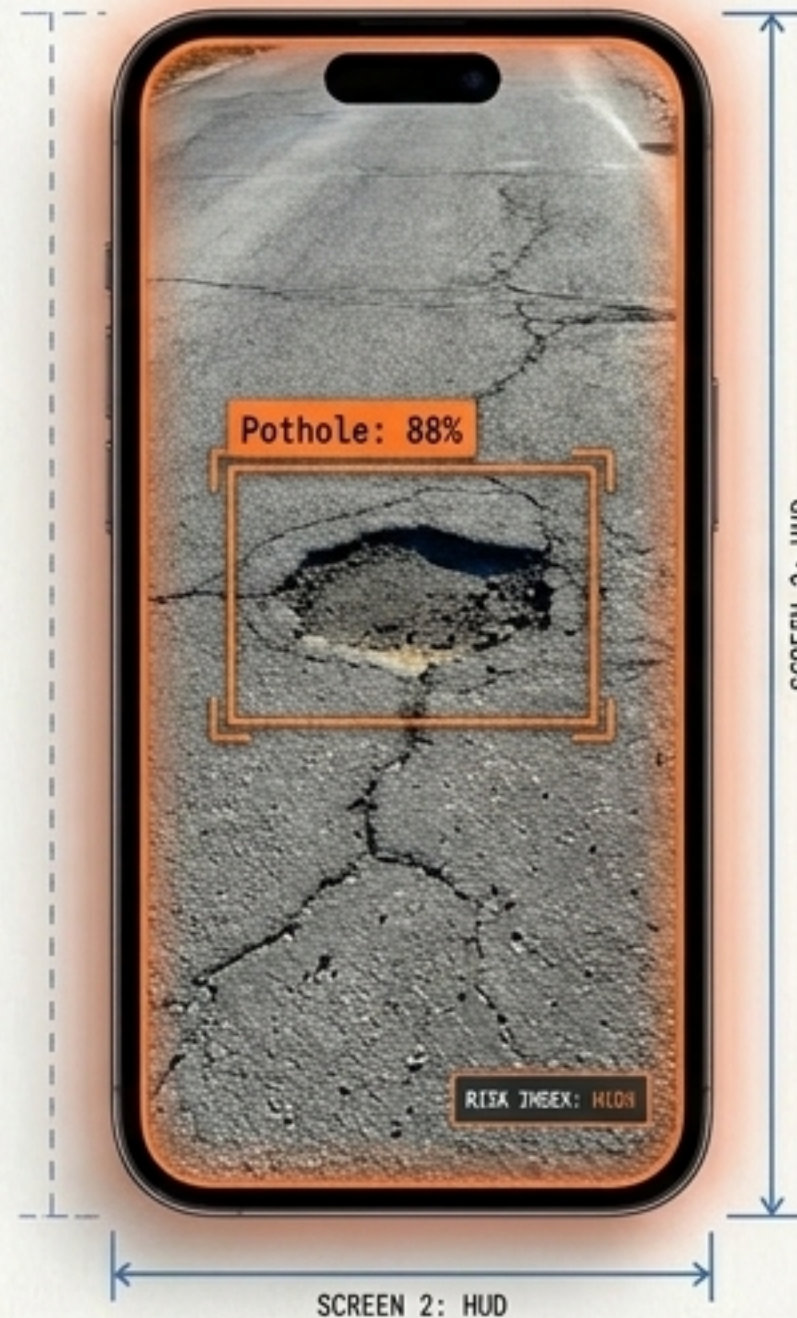
An interface engineered for the handlebar.

Sunlight-readable. Non-intrusive. Variable-frequency audio beeps allow riders to keep eyes on the road.

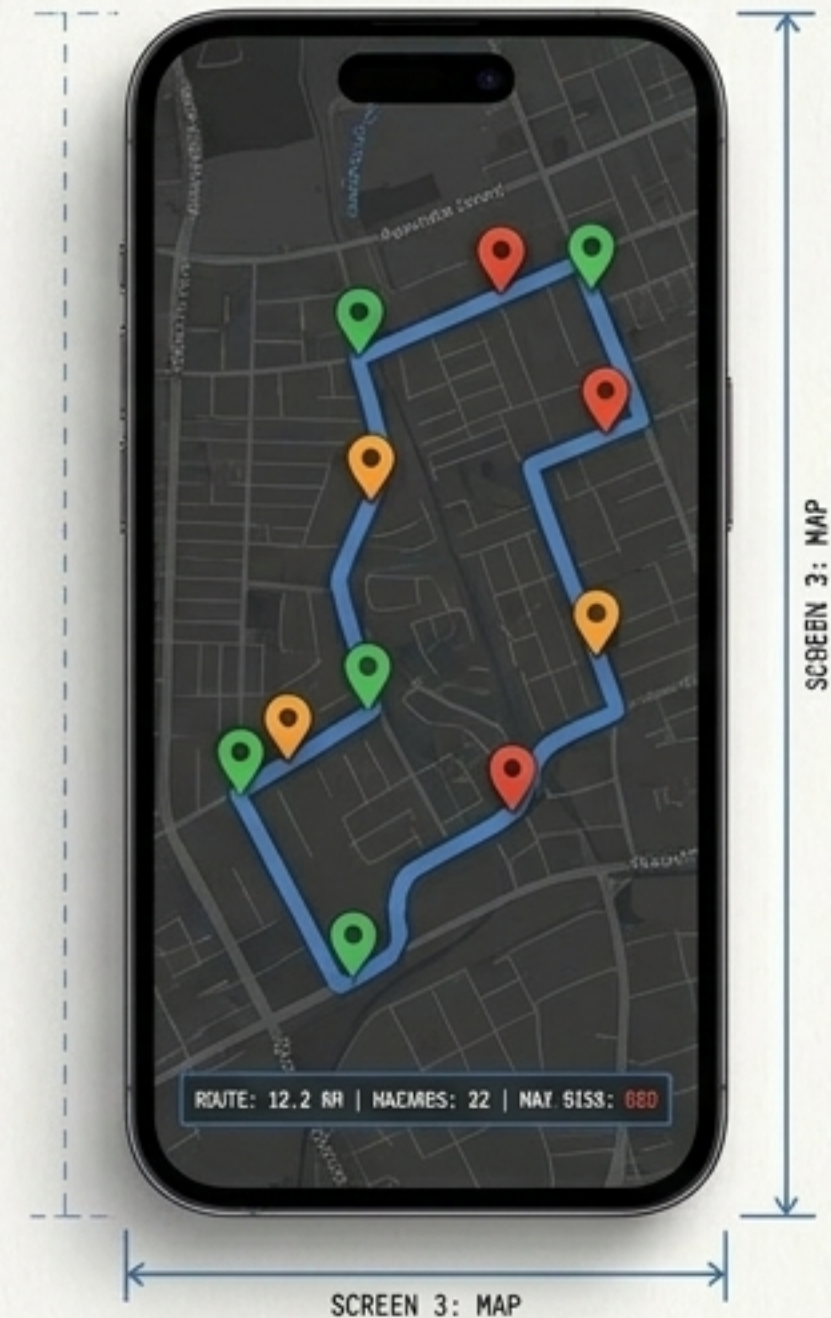
Pre-Ride Dashboard



Active HUD



Post-Ride Route Map



Validation & Performance Metrics

Definitive black-box functional testing results on standard mobile CPUs.

>15 FPS

Real-time inference executed smoothly without frame drops.

<200 ms

Total pipeline latency from image capture, to TFLite inference, to the visual red screen alert.

95.8%

Peak detection accuracy (mAP at IoU=0.5) ensuring minimal false negatives for critical hazards.

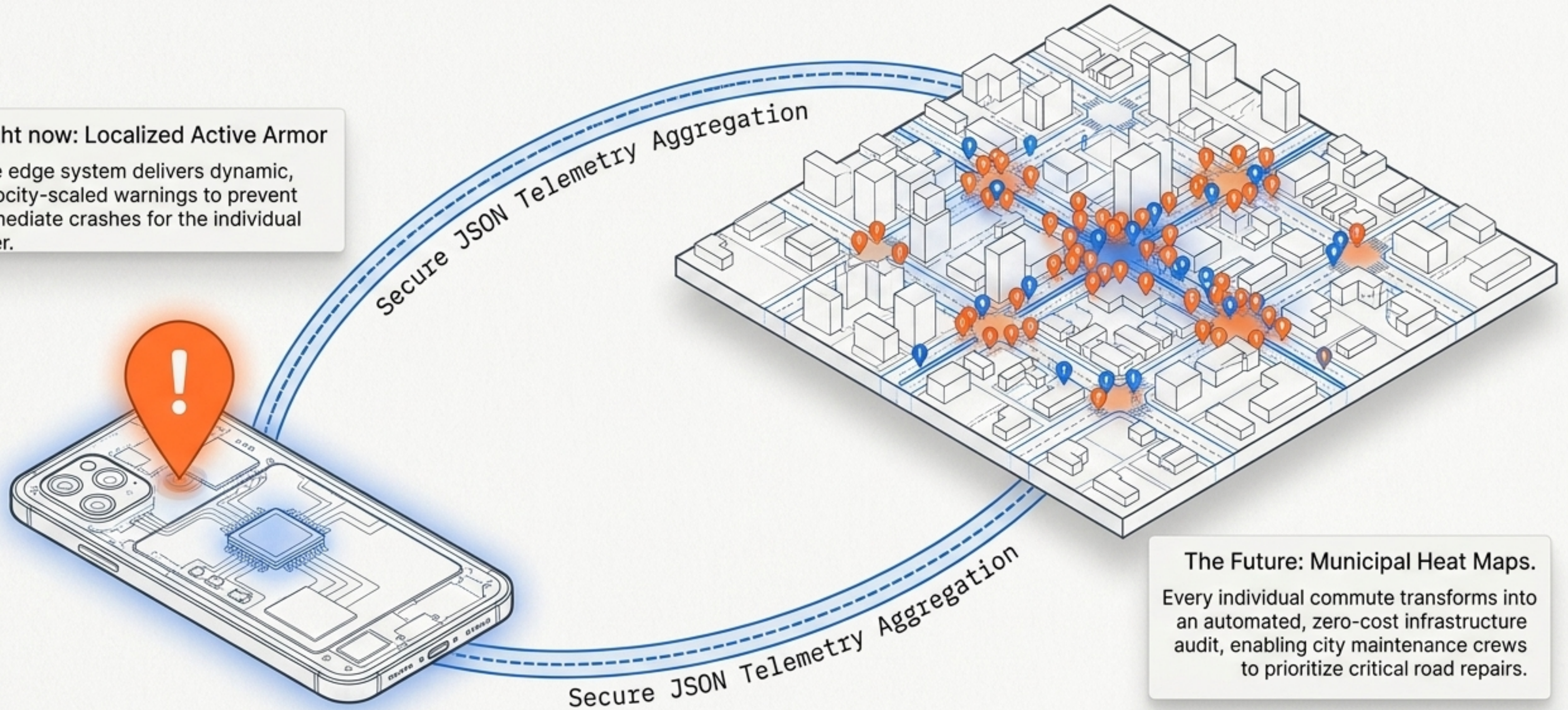
0

Total network dependency. The entire pipeline operates with 100% offline reliability and zero file-write failures.

From personal active armor to crowdsourced infrastructure audits

Right now: Localized Active Armor

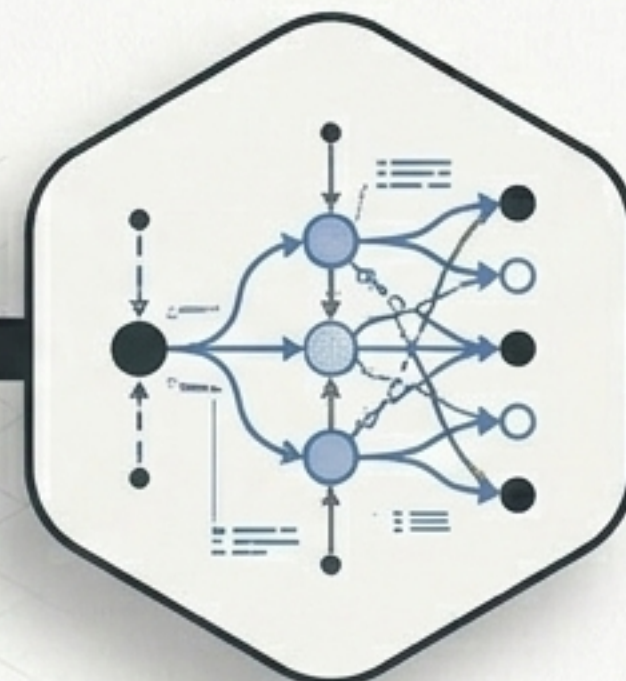
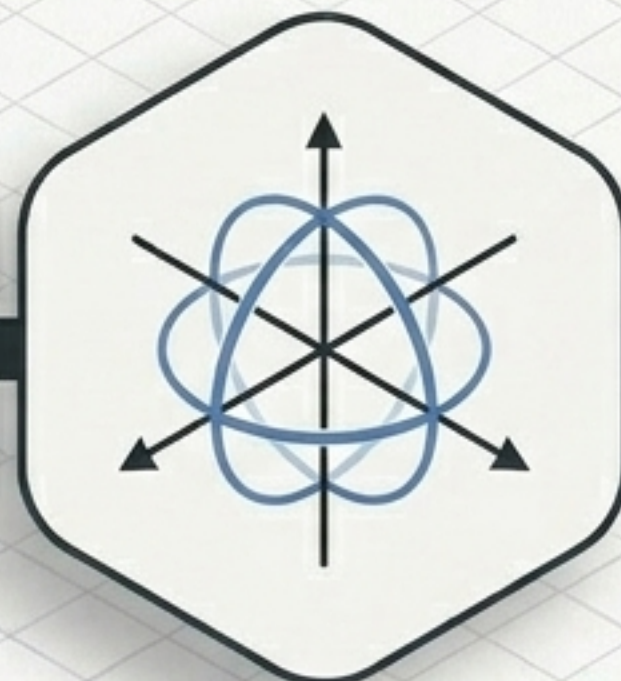
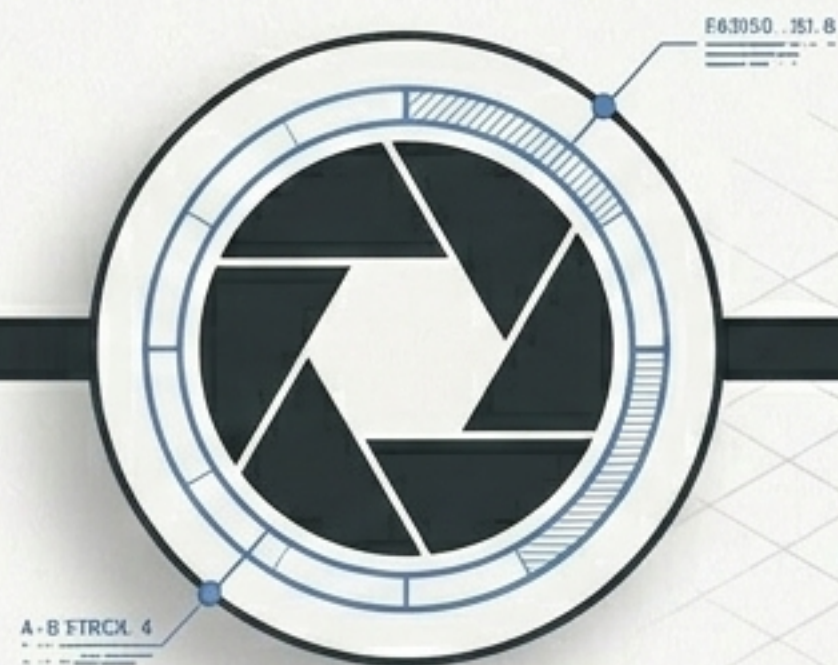
The edge system delivers dynamic, velocity-scaled warnings to prevent immediate crashes for the individual rider.



The Future: Municipal Heat Maps.

Every individual commute transforms into an automated, zero-cost infrastructure audit, enabling city maintenance crews to prioritize critical road repairs.

The horizon of micro-mobility safety.



Horizon 1: Multi-Class Vision

Expanding the dataset beyond potholes to detect lateral cracks, surface debris, loose gravel, and wet leaves. Integrating dynamic exposure control for low-light night riding.

Horizon 2: Sensor Fusion

Correlating TFLite camera vision with real-time smartphone accelerometer and gyroscope data. Verifying vertical impacts will eliminate false-positive visual alerts.

Horizon 3: Ecosystem Expansion

Implementing CoreML and TensorFlow Lite Swift bindings to expand from Android to iOS, scaling the crowdsourced network to a global user base.